MIL-P-556F
2 November 1983
SUPERSEDING
MIL-P-556E
19 August 1975

MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL: SUMP, PNEUMATIC-MOTOR-

DRIVEN, 210 GPM AT 25 FOOT TOTAL HEAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight pneumatic-motor-driven mp pumps.

1.2 Classification. The sump pumps shall be the following classes, as specified (see 6.2):

Class 1 - Regular.
Class 2 - Nonsparking.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
TT-V-119	- Varnish, Spar, Phenolic-Resin.
WW-C-633	- Coupling Halves, Cap and Wye; Quick Disconnect Pneumatic Hose, Two-Lug Universal Type.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair Parts, Accessories, and Kits, Mechanical, Packaging of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-C-4109	- Coupling Halves, Low Pressure, Air Hose, Quick-Disconnect.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-B-16541	- Bronze, Valve: Castings.
MIL-T-22085	- Tapes, Preservation and Sealing.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52404	- Connections: Hose, Fire and Water.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing: With Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1474	- Noise Limits for Army Material.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 47	- Malleable Iron Castings.
ASTM A 48	- Gray Iron Castings.
ASTM A 268	- Seamless and Welded Ferritic Stainless Steel Tubing for General Service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

ASTM A 276 - Stainless and Heat Resisting Steel Bars and Shapes.
ASTM A 536 - Ductile Iron Castings.
ASTM A 582 - Free-Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished.
ASTM B 584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE (HI)

Standards of the Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, Cleveland, OH 44107.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The sump pump shall consist of a centrifugal pump, a pneumatic motor, and the necessary connections.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Material deterioration and control. The sump pump shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to contracting officer or designated representative.

3.4 Threads. All screw threads shall be in accordance with FED-STD-H28.

3.5 Storage conditions. The sump pump shall not be damaged by storage in any ambient temperature from plus 155° F to minus 65° F, without solar radiation, for 1 hour when tested as specified in 4.5.2.4.

3.6 Safety. All rotating parts shall be guarded to prevent accidental contact by personnel. Guarded devices shall not impair the operating function.

3.6.1 Noise. The noise level at the operator position of the sump pump shall be within 85 dB(A) when tested as specified in 4.5.2.5.1. If the noise level exceeds 85 dB(A) due to exceeding the present state-of-the-art, the distance from the sump pump at which 85 dB(A) is met shall be determined. That distance shall be shown on a caution plate mounted on the sump pump, stating "CAUTION, HEARING PROTECTION REQUIRED WITHIN _____ FEET WHEN EQUIPMENT IS OPERATING". The caution shall be legible from the distance shown on the plate. The caution plate shall be colored yellow and black in accordance with MIL-STD-1474. If the caution plate is required, manuals shall include appropriate discussion of the noise hazard in accordance with MIL-STD-1474.

3.7 Reliability. The specified mean-time-between-failure (MTBF) shall be 100 hours when the sump pump is tested as specified in 4.5.2.6.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.4 and 6.5), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.7.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

3.8.2 Maintenance ratio. The sump pump shall have a maintenance ratio of not more than 0.05. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.9 Performance. Based on an air consumption rate as specified for the pneumatic motor, the sump pumps shall deliver the capacities specified in figure 2.

3.10 Pump. Pump shall be the volute centrifugal type. The pump impeller shall be mounted on the extended pneumatic motor shaft or shall be provided with an impeller shaft coupled to the motor shaft by means of a coupling. The pump and pneumatic motor shall be contained in a single housing or shall be contained in separate housings bolted together to form a single unit. All moving parts shall be enclosed. The suction inlet shall be located in the base of the pump and shall be provided with a strainer. The discharge opening of the pump shall be internally threaded 2-1/2-inches NPT. The discharge opening shall be provided with a brass pipe-to-hose adapter conforming to MIL-C-52404, type XVI, class D, having 2-1/2-inch external NPT threads on one end and 2-1/2-inch external NH threads on the other end. The pump shall operate in a dry sump for not less than 30 minutes without damage.

3.10.1 Hydrostatic pressure. The pump shall withstand an internal hydrostatic pressure of not less than 2 times the total head at shutoff conditions for not less than 5 minutes without evidence of leakage, rupture, or permanent deformation.

3.10.2 Pump impeller. The pump impeller shall be of the removable semiopen or enclosed-vane nonclogging type.

3.10.2.1 Impeller balance. The impeller shall be balanced within an out-of-balance tolerance of not more than 0.25 inch-ounce (in-oz).

3.10.3 Pump shaft.

3.10.3.1 Impeller shaft. When a separate impeller shaft is provided, the shaft shall be corrosion-resistant steel conforming to ASTM A 276, UNS S42000, or ASTM A 582, UNS S41600, cold-finished, tempered condition. The shaft shall have a Brinell hardness of not less than 225.

3.10.3.2 Extended motor shaft. When the pump impeller is mounted on the extended motor shaft, the motor shaft shall be corrosion-resistant steel conforming to ASTM A 582, UNS S41600, or carbon steel shaft with corrosion-resistant sleeve made of steel conforming to ASTM A 268, UNS S41000, or carbon steel shaft with cadmium, chrome, or nickel plating not less than 0.0001 inch thick. After fabrication, the shaft shall have a Brinell hardness of not less than 225.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

3.10.4 Bearings. The motor and impeller shaft, when provided, shall be supported by radial and thrust lubricated-for-life antifriction bearings conforming to FF-B-171, type and class optional. The bearing between the motor and the impeller shall be sealed on both sides to prevent entrance of liquids and foreign particles. The bearing seals shall be unaffected by the pumped liquids.

3.10.5 Pump-shaft seal. The pump shall be provided with a spring-loaded lubricated seal to prevent leakage between the pump impeller and the motor. The seal shall be self-alining and shall be removable from the pump shaft without disassembling the pneumatic motor unit. All components of the seal shall be resistant to corrosion.

3.10.6 Strainer. The strainer shall be copper conforming to ASTM B 584, UNS C83600, or bronze conforming to MIL-B-16541, except that class 1 sump pump may be furnished with a malleable iron strainer conforming to ASTM A 47, grade optional. The strainer shall prevent the passage of solids larger than 3/8 inch in diameter. The total area of the strainer openings shall be not less than 2.5 times the area of the pump impeller suction inlet. The bottom of the strainer shall be protected by a metal plate, which may be a separate plate or an integral part of the strainer. Wire mesh strainers, when used, shall have the top and bottom wire ends protected by a metal skirt.

3.11 Pneumatic motor. The pneumatic motor shall be the rotary-vane type. Air consumption at 80 pounds per square inch gage (psig) shall be not more than 140 cubic feet per minute (cfm). The no-load speed in revolutions per minute shall not exceed the air motor manufacturer's recommendations.

3.11.1 Air inlet. The air inlet of the motor shall be equipped with a pipe-to-hose adapter and a 3/4-inch pneumatic hose coupling conforming to WW-C-633, type III, or MIL-C-4109, type I. The air inlet shall be internally threaded NPT and shall have a zinc-free removable 24-to-60 mesh strainer.

3.11.2 Air exhaust. The air exhaust of the motor shall be 1-1/4-inch internally threaded NPT and shall be provided with a 1-1/4-inch-NPSH pipe-to-hose adapter.

3.12 Lubrication.

3.12.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable quick-disconnect fasteners may be removed to provide accessibility.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

that are subject to deterioration when exposed to climatic and pumping assembly

3.12.2 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.12.3 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with grease as follows:

(a) MIL-G-23827 - for ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as mechanical control systems; for use on gears and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

(b) MIL-G-10924 - For lubrication and surface corrosion protection of all automotive and artillery equipment operated over the temperature range of minus 65° F to plus 225° F. Not for use on machinery which comes in contact with foods or potable water.

The sump pump shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.13 Handling attachments. Each sump pump shall be provided with one or more handling attachments that will enable the sump pump to be lifted in its normal operating position. Handling attachments shall permit operating personnel wearing heavy arctic mittens to grip the attachments. Each attachment shall carry not less than 2.5 times its share of the total load imposed on the attachments. The attachments shall be attached to members which shall withstand stresses in the amount specified for the attachment. Where required, reinforcements shall be used to meet the requirements specified (see 4.5.2.10).

3.14 Identification marking. The sump pump shall be identified in accordance with MIL-STD-130. The marking shall be applied to the sump pump on plates conforming to MIL-P-514, type I, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.15 Instruction plates. Each sump pump shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the sump pump. Instruction plates shall conform to MIL-P-514, type III, composition C, or type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

3.16 Treatment and painting. The portions of the sump pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.17 Class 1, regular. The class 1 sump pump shall pump water, sewage, and sludge when completely submerged. The weight of the pump unit shall be not more than 70 pounds.

3.17.1 Pump case and impeller. The pump case and impeller shall be gray iron castings conforming to ASTM A 48, class 30A or 35A, or ductile iron castings conforming to ASTM A 536, grade 60-40-18 or 65-45-12. Impellers cast from corrosion-resistant steel are acceptable.

3.18 Class 2, nonsparking. The class 2 sump pump shall pump hydrocarbons having specific gravities from 0.68 to 0.935, at a temperature from plus 40° F to plus 80° F. The weight of the pump unit shall be not more than 80 pounds.

3.18.1 Pump case and impeller. The pump case and impeller shall be copper conforming to QQ-C-390, alloy 922, or MIL-B-16541.

3.19 Workmanship.

3.19.1 Castings and forgings. All parts, components, and assemblies of the ump pump which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, flux, and blowholes. Rework s[illegible]ll be limited to procedures which do not reduce mechanical properties or [illegible]ect function.

3.19.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.19.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.18 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the sump pump shall be at the option of the contractor.

Source: https://assist.dla.mil -- Downloaded: 202640-16T07:46Z
Check the source to verify that this is the current version before use.

3.7.5.2 Outlet connector. The pump[illegible] ... [illegible] ... [illegible] ... [

3.19.4 Welders.
by this specificatior
certification that th
either of the followi
performed and that su
code:

ANSI/AWS D1.1, Str
ANSI/ASME Boiler a

Contractors who make
position welding". S
standard welder quali
vided that the contra
contractor shall be r
operators are capable
codes.

3.19.5 Bolted con
have the burrs remove
Matching thread areas
sufficiently s! g t
shall be correctly to
protrude not more tha

4. QUALITY ASSURA

4.1 Responsibilit
tract or purchase ord
all inspection requir
in the contract or pu
facilities suitable f
herein, unless disapp
to perform any of the
spections are deemed
scribed requirements.

4.1.1 Component a
insuring that compone
in accordance with re

4.2 Classificatio
in are classified as

a. First artic

b. Quality con

c. Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Examination. The first article sump pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article sump pump shall be subjected to tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination.

4.4.2.1 Individual. Each sump pump shall be examined for the critical defect specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each sump pump shall be subjected to tests marked "X" column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to tests marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sump pump shall be examined as specified herein for the following defects:

Critical

1. Holes in casting.

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Threads not as specified.
106. Impeller shaft not as specified.
107. Extended motor shaft not as specified.
108. Bearings not as specified.
109. Pump-shaft seal not as specified.
110. Strainer not as specified.
111. Motor air inlet not as specified.
112. Motor air exhaust not as specified.
113. Lubrication fittings not as specified.
114. Lubricants not as specified.
115. Pressure-release device missing.
116. Pump case not as specified.
117. Pump impeller not as specified.
118. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Instruction plates missing, incomplete, or illegible.
203. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Conduct the tests in accordance with the applicable provisions of the test code for centrifugal pumps of the Standards of the Hydraulic Institute. Correct all test data to standard sea level atmospheric conditions of 29.92 inches of mercury and the specific weight of water of 62.3 pounds per cubic foot at a temperature of 68° F. Prior to testing, lubricate the sump pump with greases specified in 3.12.3. Conduct the storage condition test prior to the performance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

4.5.2.2 Test schedule.

TABLE I - Test schedule

First Article	Quality conformance: Individual	Quality conformance: Sample	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	X	Test conditions	4.5.2.1	-
X	X		Preassembly	4.5.2.3	3.10.2.1
X			Storage conditions	4.5.2.4	3.5
X			Safety	4.5.2.5	3.6
X			Noise	4.5.2.5.1	3.6.1
X			Reliability	4.5.2.6	3.7
X			Maintainability	4.5.2.7	3.8
X	X		Hydrostatic Pressure	4.5.2.8	3.10.1
			Performance:		
X		X	Class I sump pump	4.5.2.9.1	3.9
X		X	Class II sump pump	4.5.2.9.2	3.9
X		X	Pneumatic motor	4.5.2.9.3	3.11
X		X	Dry sump	4.5.2.9.4	3.10
X		X	Operational	4.5.2.9.5	3.9
X			Handling attachments	4.5.2.10	3.13

4.5.2.3 Preassembly. Prior to assembly, determine the pump impeller balance. An out-of-balance condition of more than 0.25 in-oz shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

4.5.2.4 Storage conditions. Completely enclose the entire sump pump in a shelter or heated container, and increase the temperature to plus 155° F. After the temperature has stabilized at plus 155° F, maintain this temperature for one hour, then lower the temperature to minus 65° F. When the temperature has stabilized at minus 65° F and this temperature has been maintained for one hour, operate the sump pump in clear water. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.5 Safety. The sump pumps shall be evaluated for safety requirements throughout testing as specified in 4.5.2. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.5.1 Noise. The sump pump shall be tested in accordance with the provisions for steady-state noise of MIL-STD-1474. Noise levels shall be taken while the sump pump is at maximum rated output. The data should include only dB(A) levels. Should the noise level exceed 85 dB(A) at any operator position, and the excess noise level can be documented as being beyond the state-of-the-art, then failure to provide the precautions specified in 3.6.1 shall constitute failure of this test.

4.5.2.6 Reliability. Using the MTBF specified in 3.7, the sump pump shall be tested as specified in 4.5.2.1 and 4.5.2.9.1 thru 4.5.2.9.5 with "accept" and "reject" criteria in accordance with figure 1. Continue testing until either an "accept" or "reject" decision is reached. For the purpose of this test, a failure is defined as any malfunction which cannot be corrected within 30 minutes by adjustment, repair, or replacement action using tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation.
b. Degradation of performance below specified levels.
c. Damage to the sump pump by continued operation.
d. Personnel safety hazards.

Nonconformance to 3.7 shall constitute failure to this test.

4.5.2.7 Maintainability.

4.5.2.7.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.8.1. Failure of the support elements to maintain the sump pump as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

4.5.2.7.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.8 Hydrostatic pressure. Subject the sump pump to an internal hydrostatic pressure 2 times the total head at shutoff conditions for a period of not less than 5 minutes. Leakage, rupture, or permanent deformation of the sump pump shall constitute failure of this test.

4.5.2.9 Performance.

4.5.2.9.1 Class I sump pump. The class I sump pump shall be placed in a sump of muddy water containing 4 pounds of clay per 100 gallons of water. Operate the sump pump and plot the head capacity curve using not less than 6 data points, of which one point shall be taken at maximum flow and one point shall be taken at shutoff conditions. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.9.2 Class II sump pump. The class 2 sump pump shall be tested the same as the class 1 sump pump, except that the class 2 sump pump shall be placed in a sump of clear water. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.9.3 Pneumatic motor. Operate the sump pump in a sump of water. The air consumption rate for any point on the curve (see figure 2) should not exceed 40 cfm when operated under 80 psig air pressure. Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.9.4 Dry sump. Operate the sump pump in a dry sump at air inlet pressure of 80 psig for a period of not less than 30 minutes. Disassemble the sump pump at completion of the test and examine it for damage. Evidence of damage to the pump shall constitute failure of this test.

4.5.2.9.5 Operational. Operate the sump pump in a wet sump at an air inlet pressure of 80 psig for a period of not less than 8 hours. An airline oiler may be used for this test. Operate the sump pump at its rated point. Disassemble the sump pump at completion of the test and examine it for damage. Evidence of damage shall constitute failure of this test.

4.5.2.10 Handling attachments. Subject each handling attachment to the load specified in 3.13. Failure or permanent deformation of either an attachment or the member to which it is attached shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

No.	Defect	A	B	Comm.
119.	Exterior surfaces not coated with preservative as required.	5.1.1.2		5.1.2
120.	Pump and air motor not coated with preservative as required.	5.1.1.3		5.1.2
121.	Pump and air motor openings not sealed as specified.	5.1.1.3		
122.	Adapters and hose couplings not coated with preservative as specified.	5.1.1.4		5.1.2
123.	Data items not preserved as specified.	5.1.1.5		
124.	Repair parts not preserved as specified.	5.1.1.6.1		5.1.2
125.	Maintenance tools not preserved as specified.	5.1.1.6.2		5.1.2
126.	Consolidation container not as specified.	5.1.1.7		
127.	Contents of consolidation container not blocked and cushioned within the container as specified.	5.1.1.7		
128.	Shipping container not as specified.	5.2.1	5.2.2	5.2.3
129.	Contents of shipping container not blocked, braced and anchored within the container as specified.	5.2.1	5.2.2	5.2.3
130.	Closure of shipping container not as specified.	5.2.1	5.2.2	5.2.3
131.	Strapping of shipping container not as specified.	5.2.1	5.2.2	
132.	Marking not as specified.	5.3.1	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

5.1.1 Level A.

5.1.1.1 Preservative applicability. The provisions of MIL-P-116 shall apply when determining the applicability of preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Exterior surfaces. Unpainted exterior ferrous metal surfaces of bolts, nuts, pins, washers and any other exterior surface requiring a contact preservative (see 5.1.1.1), shall be coated with type P-1 preservative. Unpainted ferrous metal information plates shall be coated with varnish conforming to TT-V-119 or MIL-V-13811.

5.1.1.3 Pump and air motor. The interior surfaces of the pump and air motor casing(s) and all parts within the casing(s) shall be coated with preservative lubricating oil conforming to P-10, type I, grade 30. The pump and motor shall be actuated during application of the preservative to assure complete coverage. Excess preservative shall be drained. Openings into the pump and air motor shall be sealed with plastic or metal caps or plugs. Alternatively, the openings shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, securely held in place with tape conforming to MIL-T-22085, type II.

5.1.1.4 Adapters and hose couplings. Surfaces of couplings and adapters requiring a contact preservative (see 5.1.1.1) shall be coated with P-1 preservative.

5.1.1.5 Data items. Data items for each pump unit (see 6.4) shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

Special note: When more than one data item (such as parts list, operation manual, maintenance manual or others) is required for each pump unit, all such items shall be preserved as a set in one method IC-1 or IC-3 bag.

5.1.1.6 Provisioning items (see 6.5).

5.1.1.6.1 Repair parts. Repair parts for each sump pump shall be preserved in accordance with the level A requirements of MIL-R-196. Parts not specifically mentioned therein shall be preserved as specified for parts of similar design and construction.

5.1.1.6.2 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

5.1.1.7 Consolidation. Provisioning items, data items, loose adapters and couplers and maintenance tools shall be consolidated into a close fitting box conforming to PPP-B-636, type CF, class weather-resistant. The contents shall be blocked and cushioned to prevent damage and to control or eliminate movement within the container. Box closure shall be in accordance with method IV of the appendix to the box specification.

5.1.2 Commercial. Each complete sump pump with provisioning items, data items, maintenance tools and loose components as applicable shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete sump pump, preserved as specified in 5.1.1, shall be packed in a box conforming to PPP-B-601, overseas type, style A or B. The pump unit and its consolidation container shall be arranged within the box in a manner to assure minimum cube with blocking, bracing and anchoring as specified in MIL-STD-1186. Box closure and strapping shall be as specified in the box specification and the appendix thereto. Strapping shall be flat and zinc coated.

5.2.2 Level B. Each complete sump pump, preserved as specified in 5.1.1, shall be packed as specified for level A (see 5.2.1) except that box shall be domestic type and the flat strapping need not be zinc coated.

5.2.3 Commercial. Each complete sump pump, preserved as specified in 5.1.2 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A and B. In addition to any special or other identification marking required by the contract or order (see 6.2), consolidation containers and shipping containers for each pump packed as specified in 5.2.1 and 5.2.2, shall be made in accordance with MIL-STD-129.

5.3.2 Commercial. Each pump packed as specified in 5.2.3 shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Class 1 sump pumps are intended for pumping water, sewage, and sludge from ditches, trenches, and coffer dams. Class 2 sump pumps are intended for pumping corrosive or highly flammable liquids, such as gasoline or fuel oil, from tanks or barges where nonsparking characteristics are required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class sump pump required (see 1.2).

c. When a first article is required for inspection and approval, and number of units required (see 3.2).

d. When the welding process will not be at the option of the contractor (see 3.19.3).

e. Degree of preservation and degree of packing required (see 5.1 and 5.2).

f. When additional marking is required (see 5.3.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Data requirements. The contracting officer should include requirements or such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each sump pump.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of sump pumps.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activity:
Air Force - 84

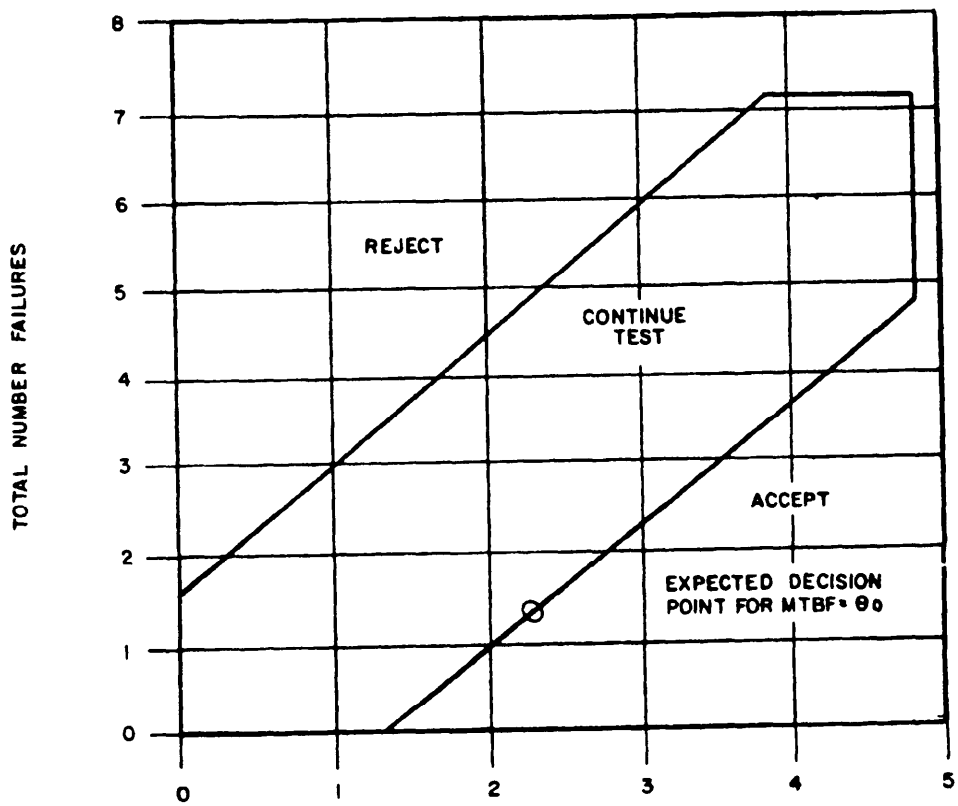
User Activity:
Navy - MC

Preparing activity:
Army - ME

Project 4320-0288

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1. Accept-reject criteria.

X-3661

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

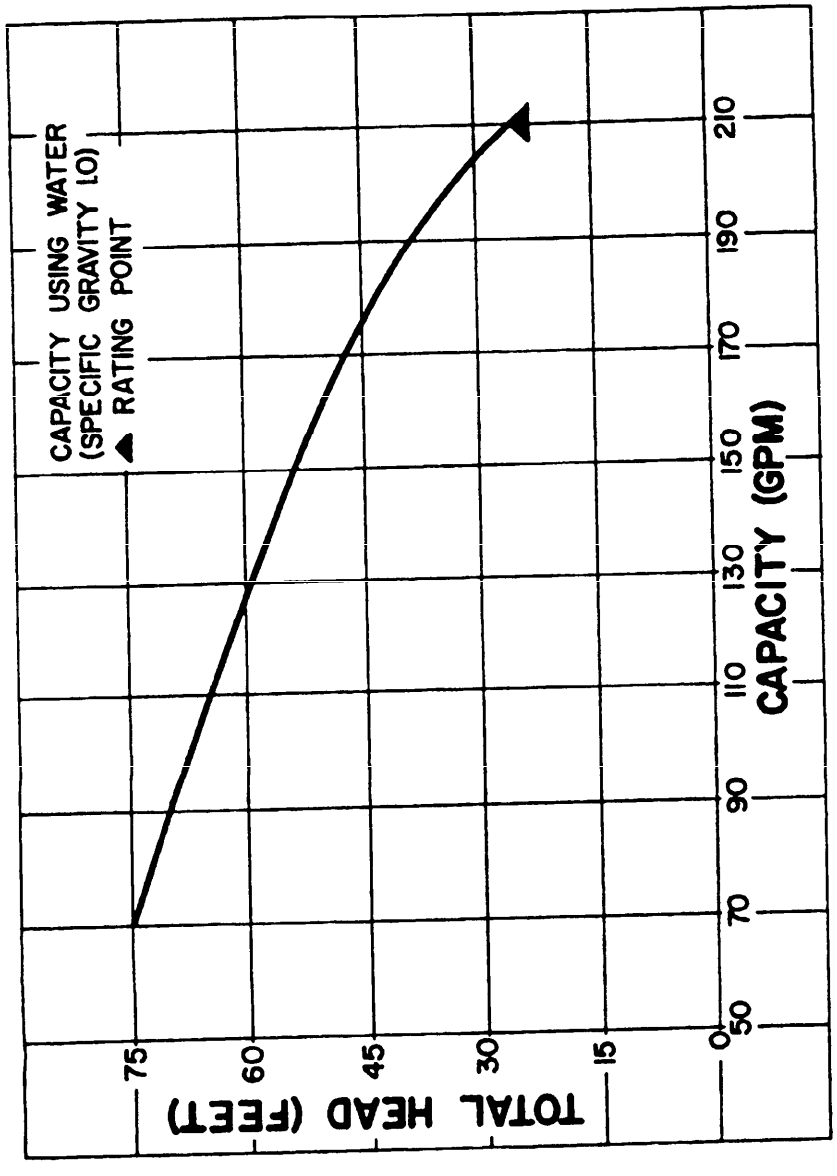


Figure 2. Pump head capacity curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-556F	Pumps, Centrifugal: Sump, Pneumatic-Motor-Driven, 210 GPM at 25 Foot Total Head

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 202610-16T07:46Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretatic was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Belvoir R&D Center
ATTN: STRBE-DS
Ft. Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:46Z
Check the source to verify that this is the current version before use.